

Pablo Lara - *Ph.D.*

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🐙 PabloELara



About me

I hold a PhD in Earth and Universe Sciences and currently work as a Postdoctoral Research Associate at Los Alamos National Laboratory (LANL). My research lies at the intersection of Machine Learning, signal processing, and geophysics. During my PhD, I developed E3WS for rapid earthquake detection and characterization; the technology is now operating in real time as part of Peru's earthquake early-warning infrastructure, contributing to a system serving more than 18 million people.

Education

- 2020 – 2024 **Ph.D. in Earth and Universe Sciences**, Université Côte d'Azur, France.
Advisors: Quentin Bletery and Jean-Paul Ampuero
Thesis title: *Detection of seismological signals based on artificial intelligence*
<https://theses.fr/s314147>
- 2018 – 2020 **Master of Science in Electrical Engineering and Computer Science**, Universidade Federal do Ceará, Brazil.
Thesis title: *Automatic multichannel volcano-seismic classification using Machine Learning and EMD*
<http://www.repositorio.ufc.br/handle/riufc/51894>
- 2017 **Electrical Engineer Certification**, Universidad Nacional de Ingenieria, Peru. Approved with honors.
Thesis title: *Design of a monitoring system for the satellite seismic network of Peru in real-time*
<http://hdl.handle.net/20.500.14076/13260>
- 2007 – 2014 **Bachelor in Science in Electrical Engineering**, Universidad Nacional de Ingenieria, Peru.

Professional experience

- Jul 2025 – Present **Postdoctoral Research Associate**, Los Alamos National Laboratory (LANL), Los Alamos, New Mexico, USA.
- Research in machine learning applied to seismology.
- 2020 – 2024 **Research Scientist**, Instituto Geofísico del Perú.
- Developed real-time volcanic event classification system for Ubinas volcano using Machine Learning.
 - Developed E3WS, the machine-learning core of the Peruvian Earthquake Early Warning System (SASPe), to detect earthquakes and estimate magnitude and hypocentral location in real time using only the first 3 seconds of P-wave data recorded by the nearest station.
 - E3WS is currently operating in real time as part of Peru's earthquake early-warning infrastructure, contributing to a system serving more than 18 million people.
 - Led the implementation, testing, and validation of the SASPe machine-learning components.

Professional experience (continued)

2014 – 2017

Software Engineer, Instituto Geofísico del Perú.

- Designed a real-time monitoring system for the National Seismic and Accelerometric Network, including:
 - a decoding system for instruments with satellite and internet telemetry,
 - a data acquisition system,
 - a monitoring platform to display the state of health of remote stations,
 - an alert system to report operational status.
- Designed intensity maps (ShakeMaps) in real time when an earthquake occurs.
- Developed software for Power Spectral Density estimation of seismic noise in real time.
- Built software to monitor servers managing the National Seismic Network in real time.

Teaching experience

2019 Winter (6 months)

Lecturer, Universidade Federal do Ceará, Sobral, Brazil. Pattern Recognition.

Research Publications

Journal Articles

- 1 **P. Lara**, H. Tavera, Q. Bletery, J.-P. Ampuero, A. Inza, D. Portugal, B. Orihuela, and F. Meza, "Implementation of the peruvian earthquake early warning system," *Bulletin of the Seismological Society of America*, vol. 115, no. 1, pp. 191–209, Dec. 2024.
- 2 A. A. T. Peixoto, C. A. R. Fernandes, **P. Lara**, and A. Inza, "Low-correlation multilinear dimensionality reduction applied to volcano-seismic classification," *Pattern Recognition*, vol. 158, p. 110 946, 2024.
- 3 **P. Lara**, Q. Bletery, J.-p. Ampuero, A. Inza, and H. Tavera, "Earthquake early warning starting from 3 s of records on a single station with machine learning," *Journal of Geophysical Research: Solid Earth*, vol. 128, no. 11, e2023JB026575, 2023.
- 4 E. Calais, S. Symithe, T. Monfret, B. Delouis, A. Lomax, F. Courboux, J. P. Ampuero, **P. Lara**, Q. Bletery, J. Chèze, et al., "Citizen seismology helps decipher the 2021 haiti earthquake," *Science*, vol. 376, no. 6590, pp. 283–287, 2022.
- 5 A. A. T. Peixoto, C. A. R. Fernandes, **P. Lara**, A. Inza, J. I. Mars, J.-P. Metaxian, M. Dalla Mura, and M. Malfante, "Tensor-based learning framework for automatic multichannel volcano-seismic classification," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 4517–4529, 2021.
- 6 **P. Lara**, C. A. R. Fernandes, A. Inza, J. I. Mars, J.-P. Métaxian, M. Dalla Mura, and M. Malfante, "Automatic multichannel volcano-seismic classification using machine learning and emd," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 1322–1331, 2020.

Conference Proceedings

- 1 A. Inza, **P. Lara**, and N. Baella, "Volcano monitoring: Hazard assessment methodology based on probabilistic analysis of ubinas volcano eruption 2006-2009," in *AGU Fall Meeting Abstracts*, vol. 2023, 2023, pp. V11D–0081.
- 2 **P. Lara**, Q. Bletery, J.-P. Ampuero, and I. Adolfo, "E3ws: Earthquake early warning starting from 3 seconds of records on a single station with machine learning," in *ERC – Tectonic Workshop*, Rome, Italy, Sep. 2023.

- 3 **P. Lara**, Q. Bletery, J.-P. Ampuero, and I. Adolfo, "Earthquake early warning with 3 seconds of records on a single station," in *EGU General Assembly 2023*, EGU23-6433, Vienna, Austria, Apr. 23–28, 2023.
- 4 **P. Lara**, Q. Bletery, J.-P. Ampuero, and A. Inza, "Earthquake early warning with 3 seconds of records on a single station," in *AGU Fall Meeting Abstracts*, vol. 2022, 2022, S53A-03.
- 5 **P. Lara**, Q. Bletery, J. P. Ampuero, A. Inza, and H. Tavera, "Earthquake early warning system based on 3 seconds of p wave: A machine learning approach for rapid detection, estimation magnitude and location," in *The physics of earthquake faulting: machine learning to illuminate earthquakes precursors and predict laboratory earthquakes*, 2021.
- 6 A. Inza, **P. Lara**, C. Alexandre, J. P. Metaxian, J. I. Mars, M. Malfante, and M. Dalla Mura, "Sismo-volcano classification in real time based on empirical mode decomposition (emd) and machine learning," in *American Geophysical Union, Fall Meeting 2020*, 2020.
- 7 **P. Lara**, A. Inza, H. Tavera, and C. A. R. Fernandes, "Efficient p-wave detection in real time for earthquake early warning system based on artificial intelligence," in *AGU Fall Meeting 2020*, 2020.

Skills

Languages	Spanish (native), English (advanced), Portuguese (advanced), French (basic).
Artificial Intelligence	TensorFlow, PyTorch, Keras, Scikit-learn.
Programming	C, C++, Python, Bash, C-Bash, HTML, PHP, Matlab, Latex.
Computer Software	MySQL, MariaDB, Linux, MongoDB, Clusters, Supercomputers.
Seismic Software	GMT, Earthworm, Seedlink, Nanometrics, Reftek, Guralp, Seiscomp3, Proxmox VM.

Awards and Honors

- 2024 **Laureate of the 2024 Innovation Trophy.** Winner in an international competition organized by the *Institut de Recherche pour le Développement* (IRD) on its 80th anniversary. Awarded at the *Science4Action* Forum in Marseille, France, recognizing pioneering interdisciplinary projects with significant global impact. <https://www.ird.fr/trophees-de-linnovation-2024-decouvrez-les-trois-laureats>
- 2020 **Doctoral Scholarship.** Awarded scholarship for doctoral thesis preparation on "Detection of seismological signals using artificial intelligence" by IRD.
- 2018 **Master Scholarship.** Received scholarship for outstanding research proposals from the Organization of American States (OAS) and the Coimbra Group of Brazilian Universities.
- 2001 **Second Place, UNI - PUCP - TRILCE - Peruvian Mathematical Society - International Commission of the Mathematics Olympics.** Achieved second place in the "National Mathematics Contest."
Second Place, TRILCE. Second place prize in the "6th Mathematics Olympiad."
- 2000 **First Place, National University of Santa.** First place prize in the "I Mathematical Logic Contest."
First Place, Editorial Active School and Ministry of Education. First place award in the "VIII Mathematics Olympiad SIGMA 2000."
- 1997 **Honorable Mention, UNASAM.** Awarded for achievements in the "V Regional Mathematics Olympiad, Chavín Region."